
AZ-Eval: A Parallel Azerbaijani–English Benchmark Reveals Negative Cross-Lingual Transfer from Kazakh Fine-Tuning

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Open multilingual models are measured on high-resource languages

Kazakh has open evaluation sets - freshqa_kazakh, defan_kazakh, BeyneleBench

Azerbaijani, to our knowledge, has none

Both are Turkic. Kazakh is written in Cyrillic, Azerbaijani in Latin.

The question

Does fine-tuning on Kazakh transfer to Azerbaijani?

Kazakh is the closest well-resourced Turkic relative of Azerbaijani. Adapting to a related language is the intuitive move for a low-resource language.

AZ-Eval - an open parallel benchmark

verified_by is a gate, not metadata: unverified rows never enter the set.

356 parallel AZ/EN short-answer items

Every item human-verified

7 categories

Accepted answers expanded

255 hand-written

101 from templates

math, history, geography, science, culture

rule-based Azeri

Four nested normalizations

STRICT → MORPH → LENIENT → TRANSLIT

suffix stripping · diacritic folding · Cyrillic→Latin

Each stage adds exactly one transformation, applied symmetrically to prediction and reference - so each score increment attributes error mass

to that single surface phenomenon.

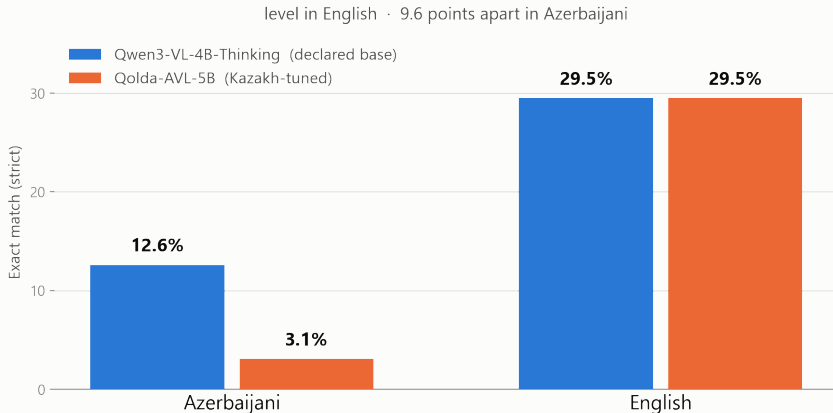
Paired comparisons · 1000-resample bootstrap · permutation test · Holm correction

Result 1 - the gap is universal

Majority-class baseline: 1.1%

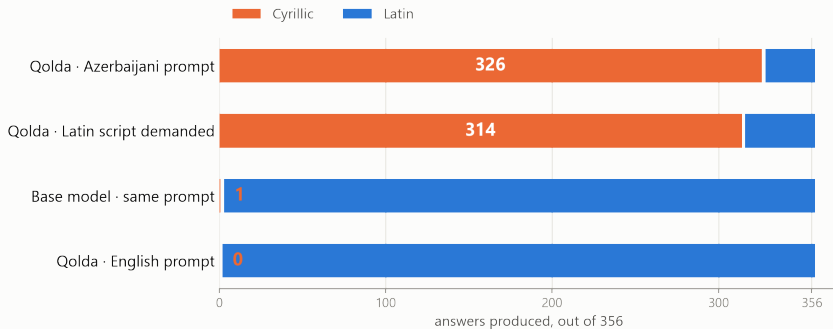
Model	AZ	EN	Gap	p (Holm)
Qwen3-1.7B	5.9%	23.6%	17.7	0.0020
Qwen3-VL-4B-Instruct	16.6%	30.6%	14.0	0.0020
Qwen3-VL-4B-Thinking	12.6%	29.5%	16.9	0.0020
issai/Qolda-AVL-5B	3.1%	29.5%	26.4	0.0020

Result 2: transfer is **NEGATIVE**



Holm-corrected $p = 0.0040$ · its own declared base, same quantization, same prompt

The mechanism is orthographic



Understands the question, finds the answer, writes it in the wrong alphabet.

Is it simply a weaker model?

Indistinguishable in English. 27.1 points apart in Azerbaijani - and transliteration closes it.

Language	Base model	Kazakh-tuned	Gap	p (Holm)
English	62.5%	59.4%	3.1	1.0000
Azerbaijani (STRICT)	37.5%	10.4%	27.1	0.0050
Azerbaijani (TRANSLIT)	38.5%	33.3%	5.2	1.0000

96 world + science items - facts every model demonstrably knows in English

Takeaway

Dataset CC BY 4.0 · Code MIT · github.com/nihatgaribli/AZ-Eval

1. Fine-tuning on a closely related language can HARM rather than help
 - when the writing systems differ.
2. Most of the damage is orthographic, not semantic
 - transliteration removes 56% of the deficit.
3. Which means it is in principle fixable
 - e.g. by constraining the output script at decoding time.

Backup slides

Backup - where the gap is widest

Mathematics: knowledge is present in English, the language blocks it.

Category	n	Qwen3-VL-4B (AZ / EN)	Qolda-AVL-5B (AZ / EN)
mathematics	28	3.6% / 25.0%	3.6% / 28.6%
world	45	51.1% / 57.8%	0.0% / 60.0%
history	45	4.4% / 0.0%	0.0% / 2.2%

Backup - what survives transliteration

Transliteration removes 56% of the deficit - but not all of it.

Residual after TRANSLIT: 5.9 points, Holm $p = 0.0364$ - still significant.

So the claim is: the damage is LARGELY orthographic, not purely orthographic.

Backup - how the dataset was built

255 items hand-written by a native Azerbaijani speaker
101 from 8 templated SPARQL queries $\times$ 4 syntactic variants
(round-robin)

No LLM authored, translated, or answered any dataset item.
Rejected candidates are retained, so the acceptance rate stays
auditable.

Items whose gold answer depends on a contested position are
excluded (6 removed).

Backup - limitations

n = 356. Confidence intervals are $\pm 2-5$ points; every reported comparison survives Holm correction.

One model pair. The 9B/8B pair did not fit in 8 GB of VRAM.

Two knowledge regimes are mixed on purpose - and kept in separate categories
so the distinction stays visible instead of averaging away.

Backup - answer extraction

Identical rules for both languages:

first line only · strip 'Cavab:' / 'Answer:' prefixes

strip markdown emphasis and trailing punctuation · remove thinking blocks

Raw model outputs are kept untouched (8 runs), so the scoring rule can be

changed and the analysis re-run without spending another GPU-hour.